

ARK-444 — PREREGISTRATION

Decision-to-Execution Integrity on IBM Quantum (Heron r2)

Remnant Fieldworks Inc. — Derek Hone

Governing principle: Proof Before Power. Prediction Before Measurement. No Rescue After Failure.

This document is the LOCK. It is committed to GitHub (derekhone/uip-phase1-testbeds, folder ark-444/) with SHA-256 hashes of all code files **before any IBM Quantum job is submitted**. The preregistration commit hash is recorded in RUN_LOG.md . Nothing below — hypotheses, arm definitions, qubit pair selection rule, shot count, metrics, or pass/fail/kill windows — may be altered after any result is observed.

Field 1 — Experiment title

Decision-to-Execution Integrity on IBM Heron hardware: a preregistered test of whether the action that executes is exactly the action that was verified, and whether a post-approval alteration of an approved action is detected and **fails closed** (the payload does not execute).

Field 2 — Experiment ID

ARK-444 (Remnant Fieldworks Inc. / Derek Hone).

Field 3 — Research question

Can the verify-then-execute (VBE) authorization boundary detect when an **approved action is altered between approval and execution**, and fail closed? Specifically: an approval commits a signature of the approved action; at execution the action's signature is re-verified and the payload fires **only if the executed action's signature matches the approved signature**. Does every class of post-approval alteration — destination changed, amount/parameter changed, operation type changed, an extra action appended, and a replayed stale approval — fail to execute the payload, while a properly **re-verified** (re-approved) mutated action correctly executes? This is a **metrological characterization of a tamper-evident decision-to-execution binding on this hardware — NOT new physics and NOT a cryptographic integrity claim** (no MAC/signature security is asserted; “integrity” here is a single-bit commitment comparison realized in a dynamic circuit).

Field 4 — Primary hypothesis H1

On the calibration-selected, connected qubit pair (Field 10), with the payload bound to a fresh execution-time verification that must match the committed approval:

- **Destination changed (Arm 2)** produces $L_{\text{corrected}} \leq 0.02$ (payload does not execute);
- **Amount/parameter changed (Arm 3)** produces $L_{\text{corrected}} \leq 0.02$;
- **Operation type changed (Arm 4)** produces $L_{\text{corrected}} \leq 0.02$;
- **Extra action appended (Arm 5)** produces $L_{\text{corrected}} \leq 0.02$;
- **Original approval replayed (Arm 6)** produces $L_{\text{corrected}} \leq 0.02$;
- **Mutated-then-reverified (Arm 7)** produces $S_{\text{reverified}} \geq 0.90$ (a re-approved action correctly executes);
- **Boundary discrimination** Δ_B between Arm 1 (approved, unchanged) and the **worst-case** altered arm is ≥ 0.70 .

Field 5 — Null hypothesis H0

Δ_B (Arm 1 vs worst-case altered arm) is statistically indistinguishable from zero and/or altered actions execute the payload at rates indistinguishable from the unchanged approved action; equivalently, the binding does not detect post-approval alteration and does not fail closed, and/or a re-verified action does not execute.

Field 6 — Secondary hypotheses

- **H2a (uniform fail-closed):** the five altered/replayed arms (2-6) are mutually statistically indistinguishable in payload activation (each $\approx$ SPAM baseline); no single alteration class leaks more than the others (descriptive; reported with 99% Wilson intervals).
- **H2b (replay $\approx$ mutation):** $P(Q_P=1 \mid \text{Arm 6 replayed})$ is statistically indistinguishable from the mutation arms (2-5) — a replayed stale approval is no better than a fresh mutation at executing the payload.
- **H2c (reverification concordance):** $S_{\text{reverified}}$ (Arm 7) is statistically indistinguishable from the unchanged approved reference S_{match} (Arm 1) within its 95% Wilson interval — re-verifying restores full execution fidelity.

Field 7 — Circuit architecture

Two-qubit dynamic circuits. Virtual $q_0 = Q_A$ (approval / integrity-commitment qubit), virtual $q_1 = Q_P$ (payload). Core mechanism — a **tamper-evident decision-to-execution binding**:

```
Approval phase:  prepare Q_A per APPROVED action ; measure(Q_A) -> ca    (committed
approval signature)
Execution phase:  reset(Q_A) ; prepare Q_A per EXECUTED action ; measure(Q_A) -> ce
(fresh execution verification)
Integrity gate:   if (ca == 1) AND (ce == 1): X(Q_P)      (payload fires iff executed
action matches the committed approval)
Payload readout:  measure(Q_P) -> cp                      (PRIMARY endpoint)
```

The payload is bound to the **fresh execution-time verification** ce , not to the stale approval ca alone. For an unchanged approved action, $ca = ce = 1$ and the payload fires. Any post-approval alteration drives $ce \neq ca$ (here $ce = 0$), so the integrity gate withholds the payload (fail closed). A replayed stale approval provides $ca = 1$ but no fresh matching verification ($ce = 0$), so it also fails closed. A mutated action that is **re-verified** re-establishes $ca = ce = 1$ and correctly executes. The single-bit commitment is a hardware abstraction of an action signature; the payload register cp is the primary readout. Named classical registers (ca , ce , cp) let SamplerV2 expose per-register counts so payload readout is unambiguous despite mid-circuit measurements. Full per-arm definitions in Field 8.

Field 8 — Arms (8 arms)

Arm	Approval (ca)	Execution verify (ce)	Integrity gate	Purpose	Endpoint
1 arm1_approved_unchanged	$ 1\rangle \rightarrow 1$	reset, $ 1\rangle \rightarrow 1$ (matches)	fires	approved action executed unchanged (reference)	$S_{\text{match}} = P(Q_P=1)$
2 arm2_destination_changed	$ 1\rangle \rightarrow 1$	reset, $ 0\rangle \rightarrow 0$ (dest $\neq$ bound)	withheld	destination changed after approval (PRIMARY)	$L_{\text{dest}} = P(Q_P=1)$
3 arm3_amount_changed	$ 1\rangle \rightarrow 1$	reset, idle $\rightarrow 0$ (param $\neq$)	withheld	amount/parameter changed after approval (PRIMARY)	$L_{\text{amount}} = P(Q_P=1)$
4 arm4_operation_changed	$ 1\rangle \rightarrow 1$	reset, $X \cdot X = I \rightarrow 0$ (op type $\neq$)	withheld	operation type changed after approval (PRIMARY)	$L_{\text{optype}} = P(Q_P=1)$
5 arm5_extra_action_appended	$ 1\rangle \rightarrow 1$	reset, $ 1\rangle$ then appended $X \rightarrow 0$	withheld	extra action appended after verification (PRIMARY)	$L_{\text{append}} = P(Q_P=1)$
6 arm6_approval_replayed	$ 0\rangle \rightarrow 0$, replay flip post-measure	reset, $ 0\rangle \rightarrow 0$	withheld (bound to original)	original approval replayed (PRIMARY)	$L_{\text{replay}} = P(Q_P=1)$
7 arm7_mutated_then_reverified	$ 1\rangle \rightarrow 1$ (fresh re-approval)	reset, $ 1\rangle \rightarrow 1$ (matches)	fires	mutated payload re-verified before execution	$S_{\text{reverified}} = P(Q_P=1)$
8 arm8_idle_spam	—	—	none	idle SPAM baseline	$\text{SPAM}_{\text{baseline}} = P(Q_P=1)$

Arms 1 and 7 share the matched (fires) construction; arms 2-6 are the fail-closed alteration/replay class (each a distinct construction that drives a commitment mismatch); arm 8 is the idle readout baseline.

Field 9 — Backend selection rule (preregistered, not cherry-picked)

Priority: (1) **ibm_marrakesh** (preferred); (2) **ibm_fez** (fallback if marrakesh non-operational or queue-prohibitive). Selected backend fixed before submission and recorded in `RUN_LOG.md`.

Field 10 — Qubit selection rule (preregistered)

Both qubits must have `readout_error < 0.020` from today's calibration snapshot; must be directly connected in the coupling map; among qualifying connected pairs, select the minimum sum of readout errors — the **same rule used in ARK-441, ARK-446, and ARK-442**. The specific pair is selected at execution time from the live calibration snapshot of the selected backend, then frozen and recorded in `RUN_LOG.md` before submission; it may not be changed thereafter. Lower physical index → Q_A (deterministic).

Field 11 — Shot count

8,192 shots per arm × 8 arms = 65,536 shots (principal job). SPAM job: 2,048 shots × 4 circuits = 8,192 shots.

Field 12 — Repetitions

1 (one principal run). No re-runs to rescue an unfavorable result (No Rescue After Failure). A protocol-deviation re-run, if ever needed, is flagged and is not primary evidence.

Field 13 — Transpiler settings

`optimization_level=1 ; initial_layout=[Q_A, Q_P]` (the selected pair); **no dynamical decoupling**; IBM Heron basis gates {cz, id, rz, sx, x}; `reset` and nested `if_test` used for the two-phase commitment. Identical settings for SPAM and principal jobs.

Field 14 — Readout mitigation decision

No readout mitigation on the primary endpoint. Raw counts are primary. The in-situ SPAM job characterizes readout error, and `L_corrected` subtracts the idle baseline analytically — but no measurement-error mitigation is applied to the primary counts. Any mitigated figures, if produced, are reported separately and labeled SECONDARY only.

Field 15 — Inclusion criteria

Principal-job results are included as primary evidence iff: (a) SPAM job completed and `SPAM_baseline ≤ 0.02` on **both** selected qubits; (b) principal job completed and returned counts for all 8 arms; (c) SPAM baseline drift between SPAM and principal jobs $≤ 0.005$; (d) no protocol deviation from this document.

Field 16 — Exclusion criteria

Exclude / flag as non-primary: backend change after submission; qubit-pair change after freeze; shot-count change; arm-definition change; calibration drift > 0.005 between jobs; any circuit alteration; job failure or partial return.

Field 17 — Metric definitions (mathematical)

```

S_match          = P(Q_P = 1 | Arm 1)           # approved-unchanged execu-
tion (reference)
L_dest          = P(Q_P = 1 | Arm 2)           # destination-changed leak-
age (primary)
L_amount        = P(Q_P = 1 | Arm 3)           # amount/parameter-changed
leakage (primary)
L_optype        = P(Q_P = 1 | Arm 4)           # operation-type-changed
leakage (primary)
L_append        = P(Q_P = 1 | Arm 5)           # appended-action leakage
(primary)
L_replay        = P(Q_P = 1 | Arm 6)           # replayed-approval leakage
(primary)
S_reverified     = P(Q_P = 1 | Arm 7)           # re-verified mutated ac-
tion execution fidelity
SPAM_baseline    = P(Q_P = 1 | Arm 8)           # idle readout error
L_worst         = max(L_dest, L_amount, L_optype, L_append, L_replay) # worst-case
altered leakage (raw)
Δ_B             = S_match - L_worst             # boundary discrimination
(Arm 1 vs worst altered arm)
L_x_corrected    = L_x - SPAM_baseline    for x in {dest, amount, optype, append, re-
play}

```

Field 18 — Statistical test

Wilson score interval for each binomial proportion. 95% CI ($z=1.96$) reported for all endpoints; 99% CI ($z=2.576$) used for leakage distinguishability diagnostics. One-sided interpretation on the primary ceilings (upper Wilson bound of each altered arm's $L_{corrected}$ vs 0.02; lower Wilson bound of $S_{reverified}$ vs 0.90).

Field 19 — Multiple-comparison correction

Bonferroni correction across the secondary hypotheses (H2a, H2b, H2c): family $\alpha = 0.05 \rightarrow$ per-test $\alpha = 0.0167$ (~98.3% CIs). The primary endpoints (H1: Arms 2-7 and Δ_B) are not penalized by the secondary family.

Field 20 — PASS condition

$L_{corrected} \leq 0.02$ for **every** altered/replayed arm (2 dest, 3 amount, 4 optype, 5 append, 6 replay) **AND** $S_{reverified}(\text{Arm 7}) \geq 0.90$ **AND** $\Delta_B(\text{Arm 1 vs worst altered arm}) \geq 0.70$, with in-situ $SPAM_{baseline} \leq 0.02$ on both qubits. A PASS means: the boundary detects every class of post-approval alteration and fails closed, while a re-verified action correctly executes.

Field 21 — FAIL condition

Any altered/replayed arm (2-6) leaks $L_{corrected} > 0.02$ while SPAM is clean ($SPAM_{baseline} \leq 0.02$); **OR** reverification fails to restore execution ($S_{reverified} < 0.90$); **OR** $\Delta_B < 0.70$ — the failure being attributable to the binding mechanism, not the qubit.

Field 22 — KILL / INDETERMINATE condition

In-situ $SPAM_{baseline} > 0.02$ on **either** qubit $\rightarrow$ INDETERMINATE / KILLED (qubit/backend disqualified); **stop immediately, submit no principal job, no rescue**. Also INDETERMINATE if SPAM baseline drifts > 0.005 between the SPAM and principal jobs (calibration drift).

Field 23 — Interpretation boundaries

Results characterize a **tamper-evident decision-to-execution binding** on this qubit pair on this backend at this calibration — a **metrological characterization, not a new physical principle and not a cryptographic integrity guarantee**. The “action signature” is abstracted to a single committed bit realized via a two-phase (approve → re-verify) dynamic circuit; the alteration arms are representative encodings of alteration classes (destination, amount, operation type, appended action) unified by the commitment mismatch they induce, plus a replayed-approval arm. “Leakage” means unauthorized payload activation (payload firing when the executed action does not match the approved action), not computational-basis leakage. A PASS demonstrates that, on this hardware and binding, post-approval alterations fail closed and a re-verified action executes; it does **not** establish security against an adversary, generalize to other bindings/qubits/backends, or assert MAC/signature strength. Findings do not generalize without replication.

Field 24 — Relationship to prior work (ARK-441 / ARK-446 / ARK-442)

ARK-441 (ibm_kingston, 2026-07-16) established the SPAM-resolved VBE authorization boundary (SPAM-corrected $L_D \leq 2\%$, $\Delta_B \geq 0.70$). ARK-446 (ibm_marrakesh) demonstrated cross-device replication (PASS). ARK-442 (ibm_marrakesh) characterized boundary degradation under verification-to-execution **delay**, plus expiry/replay/reverification (PASS). ARK-444 extends the boundary from whether an authorization is valid to whether the executed action is exactly the approved action — adding a fresh execution-time verification bound to the payload and testing five post-approval alteration classes and a reverification-recovery arm. The DENY-arm ceiling ($L_{corrected} \leq 0.02$), the reverification floor ($S_{reverified} \geq 0.90$), and the discrimination floor ($\Delta_B \geq 0.70$) carry over from ARK-441/442.

Field 25 — Exact timestamp

Preregistration authored: **2026-07-16** (UTC). Effective lock time = the GitHub preregistration commit timestamp recorded in `RUN_LOG.md`.

Field 26 — Calibration snapshot reference

A calibration snapshot of the selected backend (Heron r2) is retrieved at execution time and committed as `calibration_snapshot_marrakesh_20260716.json`; its SHA-256 is recorded in `MANIFEST.txt`. The selected qubit pair (Field 10) is derived from this snapshot and frozen in `RUN_LOG.md` before submission.

Field 27 — Ordering constraint (hard)

1. Commit this preregistration + all code files + `MANIFEST.txt` (Field 28 hashes) to GitHub. Record commit hash (the preregistration lock).
2. Retrieve live calibration → select and freeze qubit pair per Field 10 → record in `RUN_LOG.md`.
3. Run in-situ SPAM job → commit `spam_results.json`.
4. Only if `SPAM_baseline ≤ 0.02` on both qubits: submit principal job → record job ID in `RUN_LOG.md` and commit **immediately**, before reading any result. If SPAM fails the gate → KILLED, stop, no rescue.
5. Retrieve raw counts → commit `raw_results.json`.
6. Run analysis → commit `proofrecord.json`, `RESULTS.md`, plots. Tag `ark-444-v1.0`.

Field 28 — SHA-256 hashes of all code files

Authoritative hashes are maintained in `MANIFEST.txt` (committed in the same preregistration commit). The manifest covers all ARK-444 code files (qubit selection, integrity circuit generator, SPAM job,

submission, retrieval, analysis), `README.md`, and `ARK_444_preregistration.md`. Hashes are auto-filled at commit time.

AMENDMENT v1.1 — Control-flow correction (2026-07-16, before any result)

Status of v1.0 principal job: `ERROR`d, zero counts produced.

What happened. The v1.0 integrity gate was implemented as a **nested** conditional (`if ca==1: if ce==1: X(Q_P)`). On submission to `ibm_marrakesh`, the principal job (`d9cmdvsjeosc73fgfk5g`, submitted 2026-07-16, committed at LOCK-child commit `e523654`) failed to execute with **IBM error code 1524 — “Some Dynamic Circuit features are not supported (classical feedforward and control flow).”** IBM Qiskit Runtime does **not** support nested `if_test` conditionals. The job returned **no measurement data**.

Correction. The integrity gate is flattened to a **single** `if_test` on one 2-bit integrity register `ci` (`ci[0]` =approval, `ci[1]` =fresh execution-verify), firing the payload iff `ci == 0b11` (both bits set). This is exactly the **single-register feedforward pattern proven to run in ARK-442** on the same backend. Mid-circuit `reset` is retained (top-level only, never inside a conditional — a supported primitive). The **measurable semantics are identical**: the payload fires iff the approval AND the fresh execution-time verification are both 1; every alteration/replay arm still fails closed by driving the execution-verify bit to 0. Register `cp` (payload readout, PRIMARY endpoint) is unchanged. Verified by real-backend transpilation: exactly 1 non-nested `if_else` and 1 top-level `reset` per arm.

Scientific-integrity statement. Because the v1.0 job produced **zero data**, this is a **pre-data technical correction**, not a post-hoc rescue of an unfavorable result. “No Rescue After Failure” governs re-running to escape a measured FAIL; here there was no measurement to rescue. No metric, ceiling, arm definition, qubit selection (`Q_A=5`, `Q_P=6`), shot count, or decision rule (Fields 17/20–22) is changed by this amendment. The errored job ID above is retained permanently in the record. The re-locked circuit generator (`ark_444_circuits.py` v1.1) and a refreshed `MANIFEST.txt` are committed **before** the corrected principal job is submitted. The corrected run is tagged `ark-444-v1.0` (first run to yield data); this amendment and the errored job ID document the full provenance.

ARK-444 Preregistration — Remnant Fieldworks Inc. / Derek Hone — 2026-07-16. No hardware job is submitted until this document and its manifest are committed. This is a metrological characterization of a tamper-evident decision-to-execution binding, not new physics and not a cryptographic integrity guarantee. No Rescue After Failure.